ChopState

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Context first. Pressure second. Execution last.

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1. Overview

Flow.42 ChopState answers one question, bar by bar: is the market **going somewhere**, or just **wiggling sideways**? When price is chopping in a tight range, ChopState marks it. When

price breaks out and starts to trend, ChopState releases.

This matters because most trading edges only work in one of those two conditions. A breakout method bleeds in chop; a mean-reversion method gets run over in a trend. Knowing which one you are in is half the battle — and it is exactly the part that is hard to judge in the heat of the moment. ChopState makes that read explicit and consistent.

It reads two things together. The first is whether price is actually **making progress** or just covering the same ground repeatedly. The second is whether recent bars are **stacking on top of each other** in a tight, overlapping range. When both say "sideways," ChopState calls it chop. A stability layer then prevents the read from flickering back and forth on every bar, and a breakout layer lets price escape cleanly once it genuinely pushes out of the range.

ChopState is a context tool. It does not tell you to buy or sell. It tells you which kind of market you are in, so the rest of your process — and the rest of the Flow.42 Suite — can act accordingly.

2. What you see on the chart

ChopState is deliberately quiet. Its only visual is the bar color:

- **Chop bars are painted gray.** While the market is sideways, each bar is tinted a muted gray so the chop stretch is obvious at a glance.
- **Trend bars keep their normal color.** As soon as ChopState releases into a trend, bars return to your usual chart coloring.

That is the whole picture. There is no panel, no clutter — just a clear gray-versus-normal distinction that you can read peripherally while you focus on price. The indicator sits on the price panel as an overlay and updates on each bar close.

Behind the scenes ChopState also publishes its state so other indicators can use it (see Section 6), but on the chart itself the gray bars are all you need to watch.

3. Core concepts

3.1 Efficiency — is price actually getting anywhere?

Imagine price travels a long, winding path during a stretch of bars but ends up almost exactly where it started. It was busy, but it went nowhere. Now imagine price covers the same total distance but in a straight line from start to finish — that is a trend.

ChopState measures this with an **efficiency** read over a lookback period. Low efficiency means lots of motion with little net progress, which leans toward chop. High efficiency means the motion is translating into directional travel, which leans toward trend. Two thresholds, **ER Low** and **ER High**, mark the boundaries: below ER Low counts toward chop, above ER High counts toward trend.

3.2 Overlap — are the bars stacking on top of each other?

The second read looks at how much recent bars **overlap** one another. In a tight sideways range, each new bar covers much of the same price ground as the bars before it — they pile up on top of each other. In a trend, each bar tends to stake out fresh territory and overlaps its neighbours far less.

ChopState scans the last several bar pairs (set by **Overlap Lookback K**) and counts how many overlap tightly, using **Overlap Threshold** as the bar for what "tight" means. To keep the read honest, two filters exclude bars that clearly are not chop: an oversized bar (large relative to recent volatility) and a bar that made a big close-to-close jump are both skipped. Overlap also only counts while efficiency stays below the **Overlap ER Ceiling**, so a quietly trending market is not mistaken for chop.

3.3 Putting it together

Each bar earns a small chop tally from these two reads. The more the efficiency and overlap evidence agrees on "sideways," the stronger the chop case. The next section explains how that tally is turned into a stable Chop-or-Trend state.

4. How the state is decided

A raw, bar-by-bar verdict would flip constantly — one quiet bar in a trend, one sharp bar in a range, and the label would bounce around uselessly. ChopState prevents that with a **stability layer** (hysteresis).

Instead of switching the moment the evidence tips, ChopState requires the evidence to **persist**. The state only flips to Chop after the chop case has held for a set number of consecutive bars, and only flips back to Trend after the trend case has held for that many bars in a row. That count is **Consecutive Bars M**. Two score levels, **Enter Score** and **Exit Score**, set how strong the evidence must be to start counting toward chop or toward trend respectively.

The effect is a state that changes when the market genuinely changes, not every time a single bar looks unusual. Raising Consecutive Bars M makes the state calmer and slower; lowering it makes it quicker but more prone to flickering. This is the single most important dial for the feel of the indicator.

5. The two breakout paths

Stability is good inside a range, but it must not trap you when price finally escapes. ChopState has two ways to release a breakout cleanly. They are alternatives — pick one.

5.1 Smart Breakout (recommended)

With **Smart Breakout** on, ChopState maintains a dynamic "box" around the sideways range while in chop and watches for price to genuinely clear its edge. Several settings shape how it behaves:

- **Box Edge Buffer (ticks)** requires price to clear the edge by a small margin, so a one-tick brush against the boundary does not count as a break.
- **Big Body Factor Patch (x ATR)** asks for a meaningful bar — a real push, not just a wick poking out.
- **Box Stickiness (x ATR)** lets the box expand a little as the range drifts, so it does not break on every minor wick.
- **Disable Overlap Bars** briefly silences the overlap read right after a break, so the fresh trend is not immediately dragged back into chop.

- **Hold Gamma** keeps the trend state alive longer after a stronger breakout, scaling the hold time to the size of the move.

This path gives the cleanest, most responsive "release" out of chop and is the recommended default for most users.

5.2 Breakout Override (fallback)

If Smart Breakout is off, ChopState uses a simpler fallback. It tracks the range highs and lows over **Breakout Lookback** bars and releases the trend when price closes beyond that range on a sufficiently large bar, set by **Big Bar Factor (x ATR)**. After the break, the trend is held for at least **Breakout Hold Bars** bars before chop can reassert.

This fallback is more basic than Smart Breakout but dependable. Use it if you prefer a classic range-break behavior or want fewer moving parts.

6. Using ChopState with the rest of the Suite

ChopState's real power shows when it gates other tools. It publishes its read so that downstream Flow.42 indicators can ask, in effect, "are we in chop right now?" and quiet their own output when the answer is yes. Inside the Suite, VolDrive already uses ChopState this way as a suppression filter — when ChopState says the market is sideways, VolDrive can hold back signals that would likely fail in a range.

The practical workflow is simple: put ChopState on the same chart, tune it until the gray stretches match the sideways periods you would mark by hand, and let the tools that read it inherit that judgment. One well-tuned chop read then improves everything that consumes it.

7. Practical setups

Quieter, more deliberate. Raise Consecutive Bars M to 4 or 5 and raise Overlap Threshold toward 0.75–0.80. The state changes less often and demands stronger proof before leaving chop. Good when you want to avoid being chopped in and out.

Quicker, more responsive. Lower Consecutive Bars M to 3 (or 2 for maximum agility) and keep Overlap ER Ceiling conservative so trend starts are not masked. The state reacts sooner at the cost of a little more flickering.

Clean breakout release. Turn Smart Breakout on, set Box Edge Buffer to 1–2 ticks and Big Body Factor Patch around 0.80, and leave Hold Gamma at 2.0. Price that genuinely pushes out of the range frees quickly and stays freed.

8. Full parameter reference

Every setting appears in the NinjaTrader properties panel, grouped as you see it there. "Default" is the value the indicator ships with.

01 Chop Logic

Setting	Default	What it does
ER Period	14	Lookback length for the efficiency read. Shorter reacts faster; longer is smoother.
ER Low	0.35	When efficiency falls below this, the bar counts toward chop. Raise it to call chop more readily.
ER High	0.60	When efficiency rises above this, the bar counts toward trend. Lower it to release into trend more readily.
Overlap Lookback K	8	How many recent bar pairs are checked for tight overlap.
Overlap Threshold	0.70	How much two bars must overlap (0–1) to count as tight. Higher demands tighter ranges.
Small ATR Factor	0.80	Bars larger than this multiple of recent volatility are excluded from the overlap check, so an oversized bar cannot masquerade as chop.
Progress Frac Max	0.25	A bar that moved more than this fraction of recent volatility close-to-close is excluded from overlap, since real progress is not chop.

Setting	Default	What it does
Overlap ER Ceiling	0.50	Overlap only counts while efficiency is below this level, so a quietly trending market is not read as chop.
ATR Period	14	Volatility lookback used for all size-based thresholds above.

02 Breakout Patch

Setting	Default	What it does
Smart Breakout	Off	Turns on the enhanced breakout from a dynamic chop box while in chop. Recommended on for the cleanest release.
Box Edge Buffer (ticks)	1	Extra ticks beyond the box edge that price must clear to confirm a break.
Big Body Factor Patch (x ATR)	0.80	The breakout bar's body must be at least this multiple of recent volatility — a real push, not a wick.
Disable Overlap Bars	2	Briefly suspends the overlap read for this many bars after a breakout, so the fresh trend is not pulled back into chop.
Hold Gamma	2.0	Extends how long the trend state is held after a breakout, scaled by the size of the move.
Box Stickiness (x ATR)	0.30	Allows the chop box to expand by up to this multiple of volatility while in chop, so it does not break on every wick.

03 Breakout Override

These settings are the fallback breakout used when Smart Breakout is off.

Setting	Default	What it does
Breakout Lookback	20	Range lookback for the fallback breakout highs and lows.

Setting	Default	What it does
Big Bar Factor (x ATR)	1.0	The breakout bar must be at least this multiple of recent volatility to validate the fallback break.
Breakout Hold Bars	3	Minimum bars to remain in trend after a fallback breakout before chop can return.

04 Hysteresis

Setting	Default	What it does
Consecutive Bars M	3	Bars in a row the evidence must hold before the state flips. The main anti-flicker control — higher is calmer, lower is quicker.
Enter Score	1	How strong the chop evidence must be (score at or above this) to start counting toward a chop state.
Exit Score	0	How weak the chop evidence must be (score at or below this) to start counting toward a trend state.

9. Suggested starting settings per instrument

These are reasoned starting points, not backtested optima. Validate on your own setup before committing capital. Only values that differ from the defaults are listed.

ES — 1-minute (RTH)

The defaults are tuned around this case. Turn Smart Breakout on for a clean release:

- Smart Breakout = on
- Box Edge Buffer = 1 tick
- Big Body Factor Patch = 0.80
- Box Stickiness = 0.30
- Disable Overlap Bars = 2

- Hold Gamma = 2.0
- everything else at default

ES — 5-minute

A slightly slower, smoother read suits the higher timeframe:

- ER Period = 14–20
- Overlap Lookback K = 10
- Overlap Threshold = 0.75
- Progress Frac Max = 0.20
- Box Stickiness = 0.25

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Start from the ES values and adjust only if needed. These markets have jerkier micro-moves, so a slightly higher Progress Frac Max (around 0.30) often helps avoid false chop during fast directional pushes.

10. Tuning recipes

To mark more chop (stricter release). Raise Overlap Threshold toward 0.75–0.80 so only genuinely tight ranges qualify. Raise Progress Frac Max toward 0.30 so more of the sideways grind is kept as chop. Raise Consecutive Bars M to 4–5 to add stability and reduce quick flips.

To mark less chop (earlier trends). Lower Overlap ER Ceiling toward 0.45 so overlap only counts when efficiency is very low, freeing earlier as price gets going. Lower Progress Frac Max toward 0.20 so the chop case builds more slowly. Keep Consecutive Bars M at 3, or 2 if you want maximum agility and can tolerate more flicker. Lower Big Body Factor Patch toward 0.70–0.80 so breakouts release on smaller real bodies.

To free breakouts faster. Turn Smart Breakout on. Use a small Box Edge Buffer of 1–2 ticks so a clean poke beyond the edge releases chop quickly. Keep Big Body Factor Patch

around 0.80 to confirm with a real body. Leave Hold Gamma at 2.0 so stronger breakouts hold the trend longer and do not immediately re-chop.

11. Troubleshooting

Too much chop. Lower Overlap Threshold or lower ER Low so the chop case is harder to make. Check that Small ATR Factor and Progress Frac Max are not so small that ordinary bars are never excluded from the overlap check.

Too little chop. Raise Overlap Threshold, raise ER Low, or lower ER High so the read leans toward chop more easily.

Breakout sticks in trend too long. Lower Breakout Hold Bars (fallback path) or lower Box Stickiness so the box does not cling to an expanding range. With Smart Breakout, a lower Hold Gamma also shortens the post-breakout hold.

State flickers between chop and trend. Raise Consecutive Bars M. This is the dedicated anti-flicker control and almost always the right fix.

12. Glossary

Chop — a sideways market that covers the same price ground repeatedly without making net progress.

Trend — a directional market where price stakes out fresh ground and makes net progress.

Efficiency read — a measure of how much net directional travel price achieved versus the total distance it covered. Low means chop; high means trend.

Overlap — how much consecutive bars cover the same price ground. High overlap across recent bars is a hallmark of chop.

Hysteresis (stability layer) — the requirement that evidence persist for several bars before the state flips, which prevents flickering.

Chop box — the dynamic range ChopState maintains around a sideways stretch when Smart Breakout is on, used to detect a clean escape.

Breakout hold — the minimum time the trend state is kept after a breakout, so a fresh trend is not immediately pulled back into chop.

Flow.42 ChopState is a context and visualization tool. It does not provide automated trade signals or trading advice. Trade less. Trade better. Trade with clarity.

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